

ADSO

Training 6

Serveis i Timers

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1. Objectiu

Ser capaços de programar un backup periòdic.

2. Abans de començar

Hauríeu de tenir fets el codi de l'script per còpies tipus snapshot de les pràctiques anteriors:

- `backup-rsync-snapshot.sh`

3. Serveis

Programeu un servei que faci backups incrementals tipus snapshot del directori /root. Feu servir el script de la pràctica anterior. A quin directori posaries el fitxer? Quin nom li posaràs?

Els serveis de administrador es col·loquen en `/etc/systemd/system/`. Un nombre descriptiu, per exemple: `backup-snapshot.service`.

Proveu que el servei funciona amb les comandes pertinents del `systemctl`

Creem l'arxiu del servei i també creem l'script.

```
GNU nano 8.4 /etc/systemd/system/backup-snapshot.service *
[Unit]
Description=Backup Incremental Snapshot del directori /root

[Service]
Type=oneshot
ExecStart=/usr/local/bin/backup-rsync-snapshot.sh

GNU nano 8.4 /usr/local/bin/backup-rsync-snapshot.sh *
#!/bin/bash

SOURCE="/root"

DESTBASE="/tmp/backups"
DATE=$(date +%Y-%m-%d_%H%M%S)
DEST="$DESTBASE/$DATE"
LATEST="$DESTBASE/latest"

mkdir -p "$DESTBASE"

if [ -d "$LATEST" ]; then
    LINK_OPTS="--link-dest=LATEST"
else
    LINK_OPTS=""
fi

rsync -av $LINK_OPTS "$SOURCE/" "$DEST/"

rm -f "$LATEST"
ln -s "$DEST" "$LATEST"
```

Primer de tot, executem el script i després iniciem el servei que hem creat i comprovem el seu estat.

```

.mozilla/firefox/vpcgvbcl.default-esr/storage/permanent/chrome/idb/3870112724rsegmnoittet-es.files/
.mozilla/firefox/vpcgvbcl.default-esr/storage/permanent/chrome/idb/3870112724rsegmnoittet-es.files/1
.mozilla/firefox/vpcgvbcl.default-esr/storage/permanent/chrome/idb/3870112724rsegmnoittet-es.files/2
.mozilla/firefox/vpcgvbcl.default-esr/storage/permanent/chrome/idb/3870112724rsegmnoittet-es.files/journals/
.mozilla/firefox/vpcgvbcl.default-esr/storage/temporary/
.ssh/
.ssh/known_hosts
.ssh/known_hosts.old
Baixades/
Documents/
Escriptori/
Imatges/
Música/
Plantilles/
Públic/
Vídeos/
backup/
backup/backup-rsync/

sent 205.859.247 bytes received 43.038 bytes 137.268.190,00 bytes/sec
total size is 205.595.523 speedup is 1,00
root@davidP (dt. de des. 09):~# systemctl start backup-snapshot.service
root@davidP (dt. de des. 09):~# systemctl status backup-snapshot.service
● backup-snapshot.service - Backup Incremental Snapshot del directori /root
   Loaded: loaded (/etc/systemd/system/backup-snapshot.service; static)
   Active: inactive (dead)

de des. 09 00:41:04 aso-client backup-rsync-snapshot.sh[12565]: Plantilles/
de des. 09 00:41:04 aso-client backup-rsync-snapshot.sh[12565]: Públic/
de des. 09 00:41:04 aso-client backup-rsync-snapshot.sh[12565]: Vídeos/
de des. 09 00:41:04 aso-client backup-rsync-snapshot.sh[12565]: backup/
de des. 09 00:41:04 aso-client backup-rsync-snapshot.sh[12565]: backup/backup-rsync/
de des. 09 00:41:04 aso-client backup-rsync-snapshot.sh[12565]: sent 205.859.247 bytes received 43.038 bytes 137.268.190,00 bytes/sec
de des. 09 00:41:04 aso-client backup-rsync-snapshot.sh[12565]: total size is 205.595.523 speedup is 1,00
de des. 09 00:41:04 aso-client systemd[1]: backup-snapshot.service: Deactivated successfully.
de des. 09 00:41:04 aso-client systemd[1]: Finished backup-snapshot.service - Backup Incremental Snapshot del directori /root.
de des. 09 00:41:04 aso-client systemd[1]: backup-snapshot.service: Consumed 832ms CPU time, 207.6M memory peak.
lines 1-14/14 (FND)

```

4. Timers per tasques periòdiques

Programem un timer que iniciï el servei anterior per fer un backup cada diumenge a les 22:00. Mostra com ho has fet i comprova que funciona correctament.

Ara per el timer, creem un altre codi per el timer. Fem un reload del daemon i iniciem el timer (enable) i comprovem amb la ultima comanda que tenim programada per el diumenge a les 22:00 el backup.

```

GNU nano 8.4 /etc/systemd/system/backup-snapshot.timer *
[Unit]
Description=Ejecuta el backup cada domingo a las 22:00

[Timer]
OnCalendar=Sun 22:00
Persistent=true

[Install]
WantedBy=timers.target

root@davidP (dt. de des. 09):~# systemctl daemon-reload
root@davidP (dt. de des. 09):~# systemctl enable --now backup-snapshot.timer
Created symlink '/etc/systemd/system/timers.target' → '/etc/systemd/system/backup-snapshot.timer'.
root@davidP (dt. de des. 09):~#

```

```
root@davidP (dt. de des. 09):-# systemctl list-timers --all
NEXT LEFT LAST PASSED UNIT ACTIVATES
Tue 2025-12-09 06:02:42 UTC 5h 15min Mon 2025-12-08 23:02:53 UTC 1h 44min ago apt-daily-upgrade.timer apt-daily-upgrade.service
Tue 2025-12-09 06:33:46 UTC 5h 46min Tue 2025-12-09 00:25:42 UTC 21min ago apt-daily.timer apt-daily.service
Tue 2025-12-09 07:31:48 UTC 6h Mon 2025-12-08 23:30:41 UTC 1h 16min ago anacron.timer anacron.service
Tue 2025-12-09 21:54:24 UTC 21h Thu 2025-10-23 07:05:29 UTC 2h 52min ago systemd-tmpfiles-clean.timer systemd-tmpfiles-clean.service
Wed 2025-12-10 00:00:00 UTC 23h Tue 2025-12-09 00:00:01 UTC 46min ago dpkg-db-backup.timer dpkg-db-backup.service
Wed 2025-12-10 00:00:00 UTC 23h Tue 2025-12-09 00:00:01 UTC 46min ago man-db.timer man-db.service
Wed 2025-12-10 00:27:00 UTC 23h Tue 2025-12-09 00:25:42 UTC 21min ago logrotate.timer logrotate.service
Sun 2025-12-14 03:10:30 UTC 5 days Mon 2025-12-08 23:02:53 UTC 1h 44min ago e2scrub_all.timer e2scrub_all.service
Sun 2025-12-14 22:00:00 UTC 5 days - - backup-snapshot.timer backup-snapshot.service
```